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Data Science

PLANORA- A Blend of Architecture And Vision

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Project Guide
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Outline

- Introduction
- Literature Survey of the existing systems
- Limitations of the existing systems
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Sustainable Development Goals (SDG) mapped

SDG 9 : Industry, Innovation and Infrastructure
This project uses Artificial Intelligence, deep learning (Mask R-CNN), and computer vision to automatically convert 2D floor plans into 3D models. It promotes technological innovation in the architecture and construction industry by providing a smarter and more efficient way to design and visualize building structures.

Introduction

- Visualization of architectural designs is essential for planning, communication, and decision-making.
- Converting 2D floor plans into 3D models is traditionally time-consuming and requires technical expertise.
- Existing tools often lack accuracy, customization, and ease of use.
- Advances in Artificial Intelligence enable faster and more automated 2D-to-3D conversion.
- The proposed system aims to provide an efficient, user-friendly solution with high-quality and customizable 3D outputs

Objectives

- To implement Mask R-CNN for accurate detection and segmentation of architectural elements such as walls, doors, and rooms from 2D floor plans.
- To apply rule-based algorithms and spatial optimization techniques to enhance layout accuracy, structural consistency, and efficient space utilization.
- To enable users to upload 2D floor plans in formats such as JPG, PNG, PDF, and SVG and automatically convert them into precise and interactive 3D models using AI-based methods.
- To provide customizable and interactive 3D visualization, including lighting, textures, material settings, and multiple viewing modes such as bird’s-eye view and walkthrough navigation.

Literature Survey of the existing system

Sr. No.	Title	Author(s)	Year	Outcomes	Methodology	Result
1	Deep Learning Models for the Recognition of Building Elements in Residential Floor Plan Drawings	N. Bróż, P. Nazarko, A. Prokop, A. Borowiec	2025	Focuses on recognizing building elements such as walls, doors, and windows from floor plan images.	Uses deep learning-based object detection and segmentation models to analyze residential floor plan drawings.	Demonstrated improved accuracy in detecting structural components from architectural floor plans.
2	A Comprehensive Survey of Floor Plan Image Analysis and Related Applications	R. Khade, K. Jariwala, C. Chattopadhyay	2025	Reviews various AI techniques used in floor plan analysis and discusses challenges in recognizing architectural layouts.	Surveys multiple approaches including deep learning, object detection, and spatial analysis methods.	Shows that deep learning models significantly improve automated floor plan analysis and layout extraction.

Sr. No.	Title	Author(s)	Year	Outcomes	Methodology	Result
3	Progressive Active Learning for Floor Plan Image Analysis	D. Al-Turki et al.	2023	Proposes an AI approach to detect important elements in floor plan images more efficiently.	Uses progressive active learning with annotated floor plan datasets to improve detection performance.	Achieves better accuracy in recognizing floor plan components while reducing manual annotation effort.

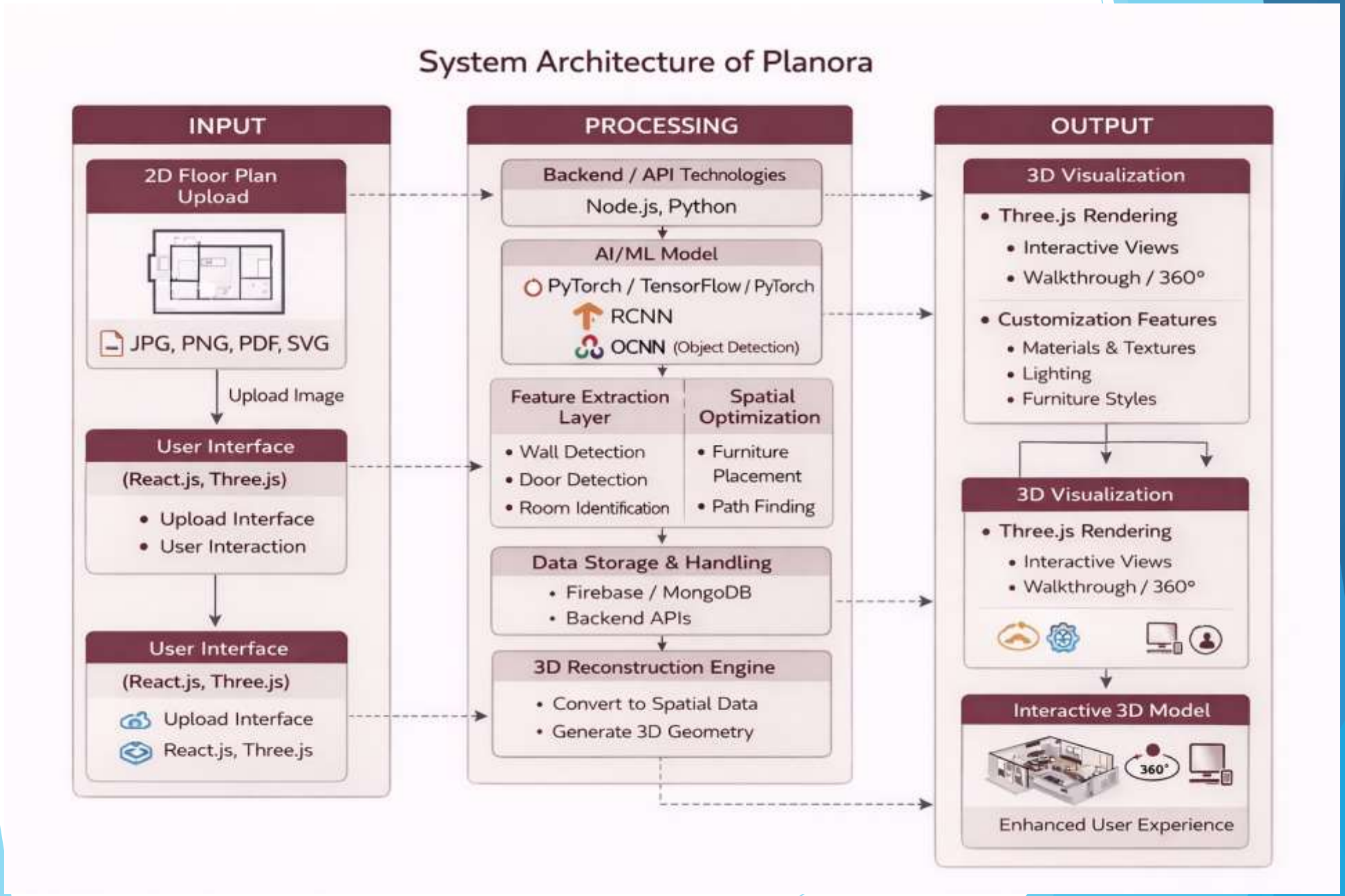
Limitations of existing systems

- Limited accuracy in converting complex or irregular 2D floor plans into precise 3D models.
- Restricted customization options for rendering styles, materials, and lighting.
- Lack of intuitive or immersive viewing modes, reducing user experience.
- High processing time or requirement for powerful hardware, making them less accessible.

Problem Statement

- Existing systems lack accuracy when converting complex 2D floor plans into reliable 3D models.
- Limited customization options for rendering styles, lighting, and materials restrict design flexibility.
- Many tools are not user-friendly and require technical expertise to operate effectively.
- Insufficient viewing options reduce the ability to fully visualize and explore the generated models.
- Low-quality outputs or high processing requirements make them less suitable for professional use.

System Design



Technologies and methodologies

- Frontend: React.js for building an interactive user interface.
- 3D Visualization: Three.js (WebGL) for rendering and displaying 3D models in the browser.
- Backend: Python (Flask/FastAPI) for handling requests and processing data
- Object Detection Algorithm: Mask-RCNN (Region-Based Convolutional Neural Network) for detecting walls, doors, and other structural elements from floor plans, Rule Based Algorithm, Spatial Optimization.
- Computer Vision: OpenCV for image preprocessing and feature extraction.
- File Handling: Support for JPG, PNG, PDF, and SVG formats
- Dataset Used: 5KCubiCasa Dataset

Implementation

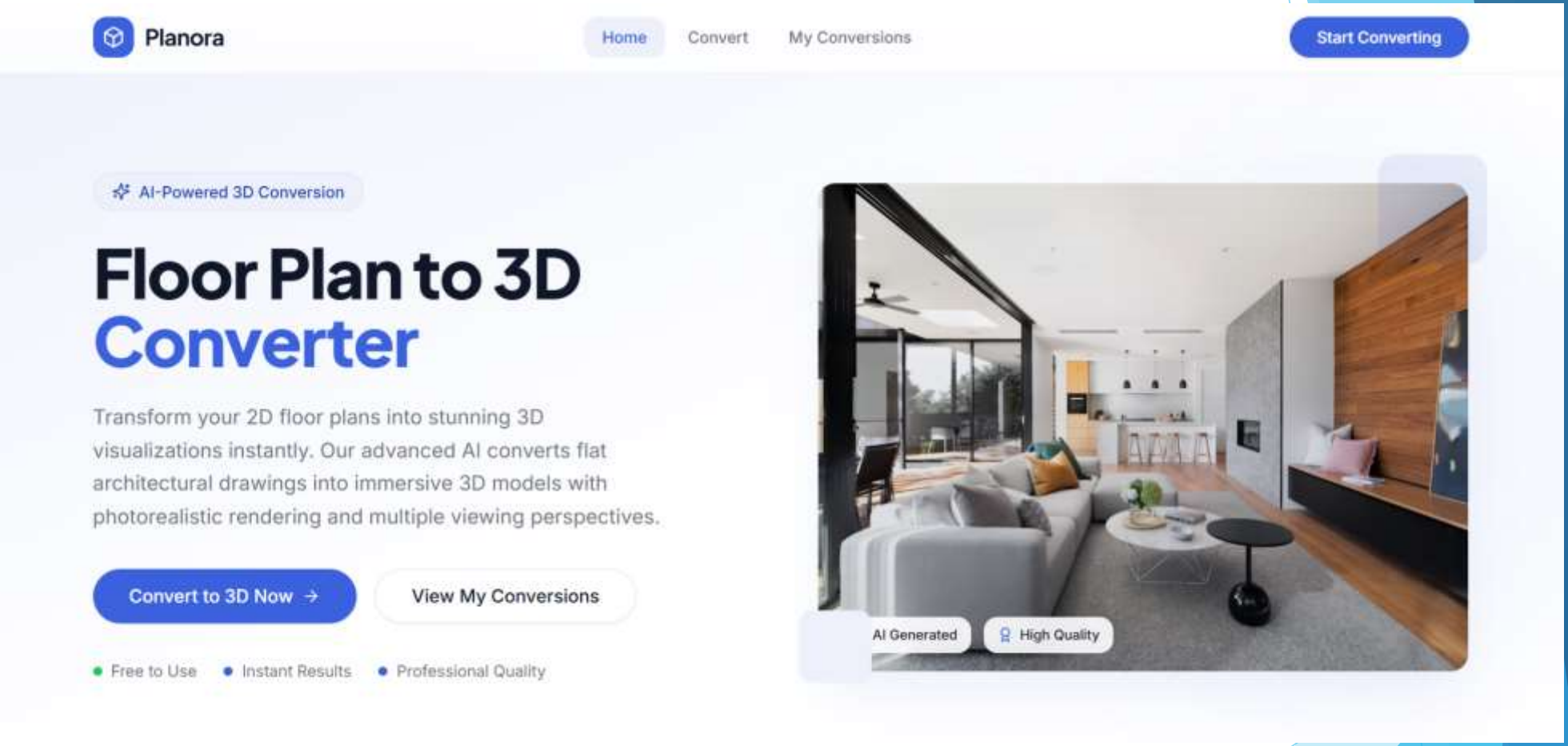


Fig 1: Home Page- Planora

Implementation

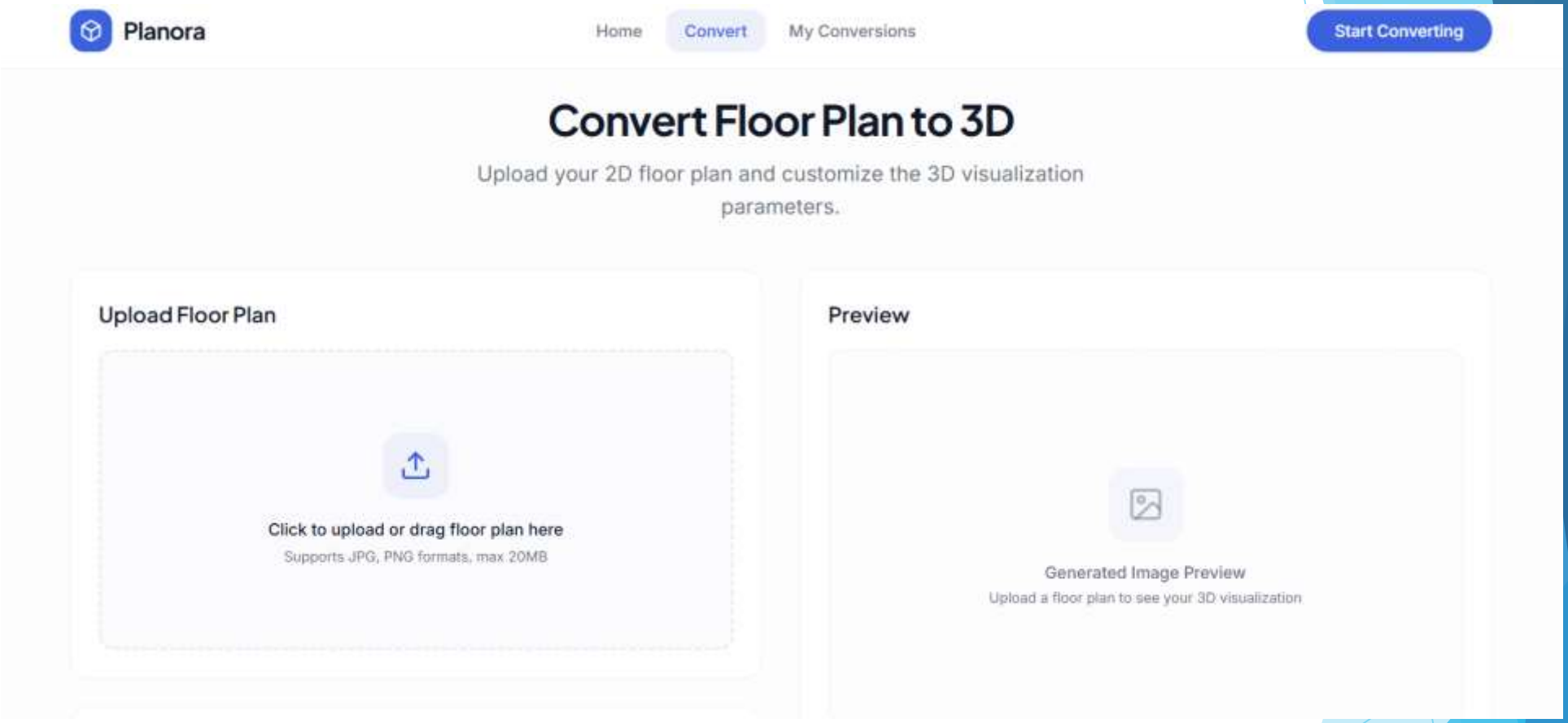


Fig 2- Conversion Page- 2D to 3D (Upload Image)

Implementation

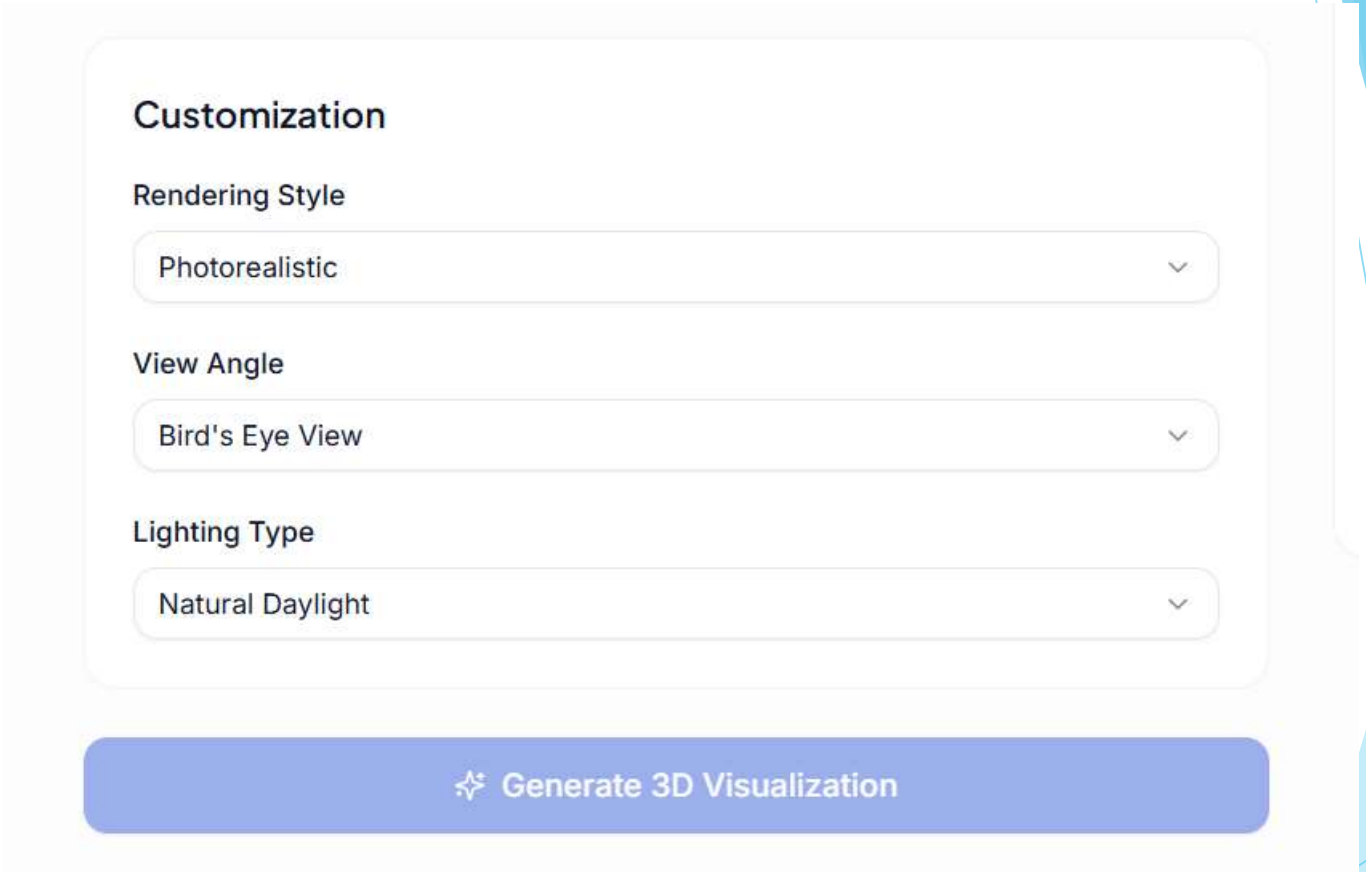
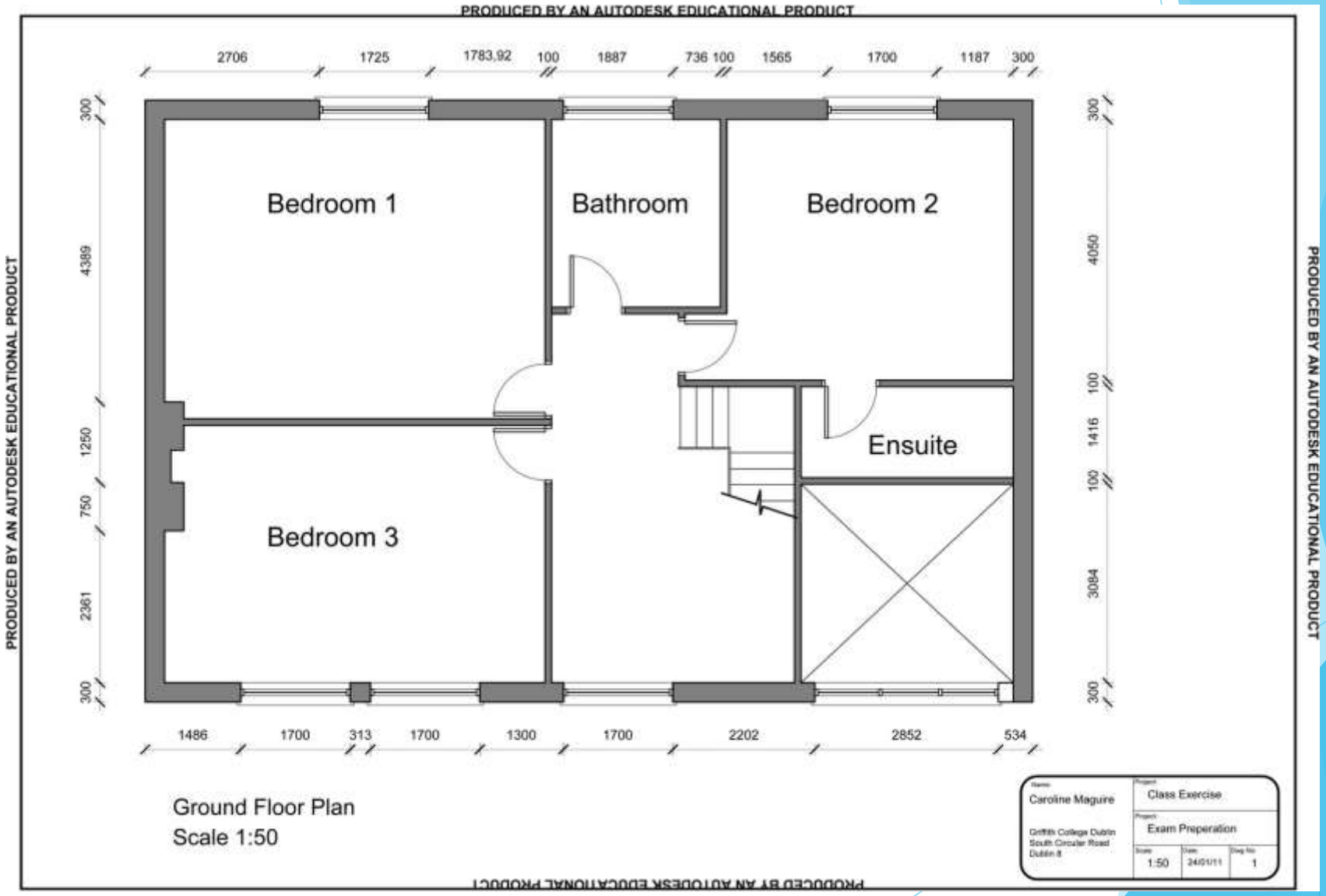


Fig 3- Rendering styles, View Angle, and Lighting Type

Implementation (2D Floor Plan)



Implementation (3D Floor Plan)

Conclusion

- The system successfully converts 2D floor plans into accurate 3D models using AI.
- It reduces manual effort and speeds up the design visualization process.
- Customization features enhance flexibility and user experience.
- Multiple viewing modes improve understanding and presentation of designs.
- The solution provides a scalable and efficient approach for modern architectural visualization

References

[1] N. Bróz et al.:*Deep Learning Models for Building Element Recognition*, IEEE Access, 2025. https://ieeexplore.ieee.org/

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Thank You...!!